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Unit “ 4 : Session “ 3 : SLO “ 2 | Odd Man Out “ Instruction Execution

■ I. Odd Man Out

1. Group Fetch, Decode, Execute, Store ‘ Store “ Fetch, Decode and Execute are the three fundamental stages of the instruction cycle; storing is a sub-step of execution (write-back).
2. Group PC, IR, ALU, CU ‘ ALU “ it is the data-processing unit; PC, IR and CU all serve instruction sequencing/control.
3. Group ADD, SUB, MOV, HALT ‘ HALT “ it stops the machine; the others are data-processing/transfer instructions that produce a result.
4. Group Register, Cache, RAM, Hard Disk ‘ Hard Disk “ it is non-volatile secondary storage; the others are fast volatile storage inside the memory hierarchy.
5. Group Zero Flag, Overflow Flag, Carry Flag, Clock Signal ‘ Clock Signal “ it drives timing; the others are ALU status flags.
6. Group Immediate, Direct, Indirect, Parallel ‘ Parallel “ it is not an addressing mode; the other three are.
7. Group RISC, CISC, Pipelining, GPU ‘ GPU “ it is a hardware device; RISC/CISC are ISA design philosophies and pipelining is an execution technique.
8. Group Branch, Jump, Call, Load ‘ Load “ it is a data-transfer instruction; the others alter the control flow.
9. Group Operand Fetch, Instruction Fetch, Write Back, Memory Access ‘ Instruction Fetch “ it fetches the instruction word; the others move data between CPU and memory.
10. Group Microoperation, Macrooperation, Nanoperation, Clock Cycle ‘ Nanoperation “ it is not a real architectural term; microoperations execute instructions and clock cycles time them.